

大宗商品分析

伊朗时局对能源价格构成的风险 (摘要)

- 通过霍尔木兹海峡的油轮运输（通常全球五分之一的石油和液化天然气供应经过该海峡）看似已严重中断，在船只受损的报道传出后，许多航运商、石油生产商和保险公司已转向谨慎的观望模式。我们估算了中东事态发展带来的上行风险。目前，我们在基本情景（假设不会出现持续的供应中断）下的能源价格预测保持不变。
- 石油。基于零售价格在周末15%的涨幅，我们估算原油价格中的实时风险溢价为18美元/桶，大致相当于我们估算的霍尔木兹海峡流量完全中断六周所带来的公允价值影响（已考虑了使用备用管道容量可产生部分抵消作用）。若仅假设50%的流量中断一个月，这一估计影响将降至+4美元。然而，如果市场因更持久的供应中断风险而要求溢价，则油价可能会大幅上涨。
- 天然气。直到上周五，欧洲天然气（TTF）和现货液化天然气（JKM）价格几乎未计入风险溢价，而我们认为，通过霍尔木兹海峡的液化天然气供应可能出现的持续中断给价格带来了显著上行风险。在流量中断一个月的假设情景下，我们认为TTF和JKM价格很可能接近74欧元/MWh（25美元/mmBtu）——较当前水平高出130%，这是2022年欧洲能源危机期间引发了大规模天然气需求响应的一个阈值。我们认为美国天然气价格面临的上行风险有限。
- 关注要点。霍尔木兹海峡的流量，以及美国、伊朗、中国和海湾合作委员会成员国就该海峡及更广泛冲突可能进行的沟通，是当前能源市场需要关注的最重要变量。

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The Risks to Energy Prices From Iran

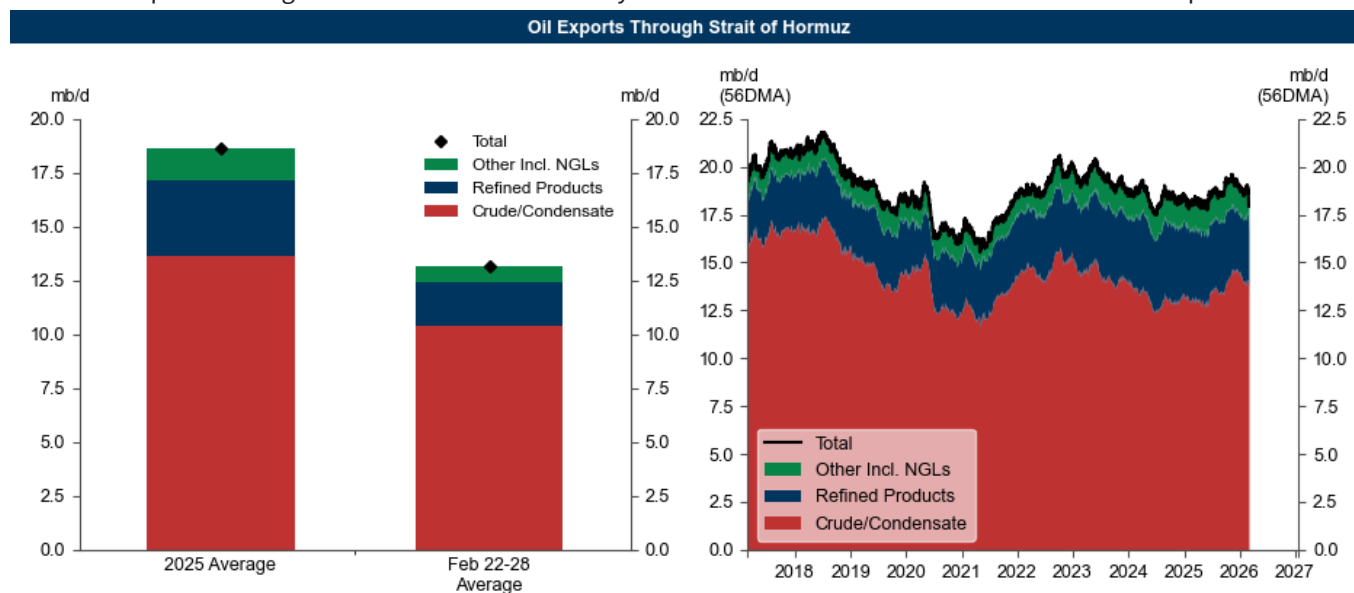
Q0. What has happened in the Middle East over the weekend?

Iran's Supreme Leader Ayatollah Ali Khamenei has been killed after Israel and the US launched an ongoing military operation against Iran's leadership and military. Iran has responded by firing ballistic missiles and drones at US assets and allies across the region, targeting Israel, Bahrain, Kuwait, Qatar, Oman, the United Arab Emirates, Saudi Arabia, and Jordan.

Q1. What is the evidence on disruptions to energy flows? Have you changed your baseline energy price forecasts?

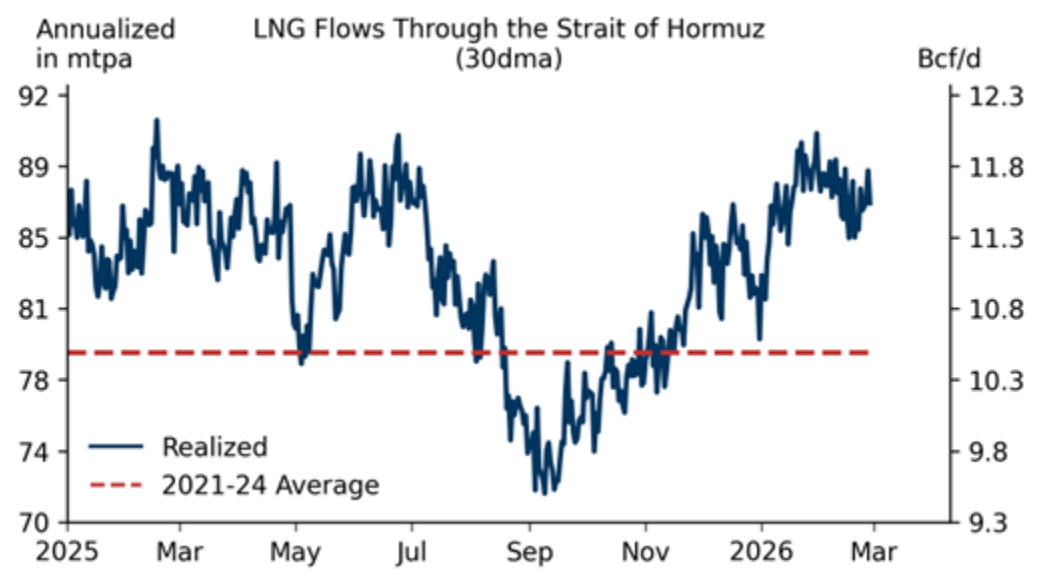
While flow data through the Strait of Hormuz through last Friday point to somewhat above annual average volumes for oil (图表 1) and for LNG (图表 2), there have been reports over the weekend of large drops in the number of tankers crossing through the Strait, and of three oil tankers reportedly damaged in the region. We think traffic may remain disrupted until there is further clarity on risks (e.g. mines, strikes). We assume no sustained flow disruptions in our baseline energy price forecasts, which we have left unchanged for now, and we will continue to monitor high frequency flows through the Strait closely.

图表 1: Oil Exports Through the Strait of Hormuz Already Decreased Last Week Ahead of the Weekend Disruptions



资料来源: Kpler, 高盛全球投资研究部

图表 2: Flows Through the Strait of Hormuz Were Normal Through Friday; Approximately 80 mtpa of LNG (19% of Global Supply), Mainly From Qatar, Flows Through the Strait of Hormuz



资料来源: Kpler, 高盛全球投资研究部

Risks to Oil Prices

Supply at Risk

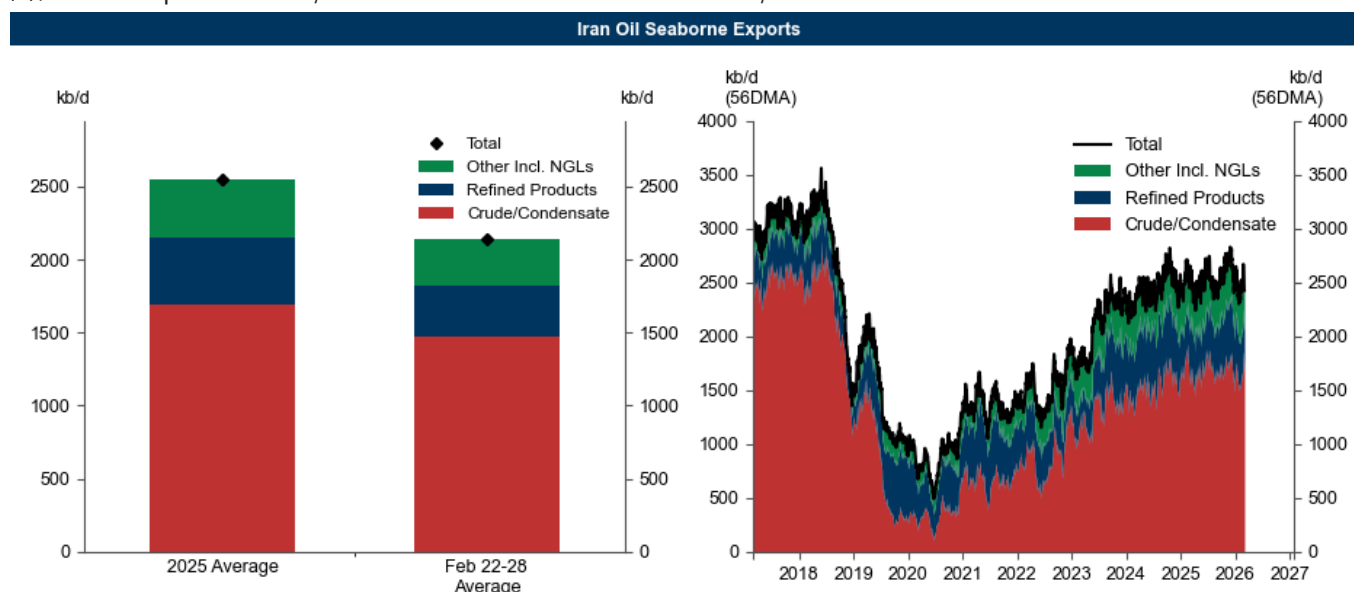
Q2. Is oil infrastructure in Iran and the Middle East damaged?

To our knowledge, there is no confirmed damage to oil production or to oil export infrastructure in the region.¹

Q3. How much oil supply from Iran and from the Middle East via the Strait of Hormuz is at risk?

Iran produced around 3.5mb/d of crude oil and about 0.8mb/d of condensate in 2025, together worth 4% of global oil supply. Iran's 2025 exports averaged 1.7mb/d for crude and condensates, 0.6mb/d for refined products, and 0.4mb/d for Natural Gas Liquids (NGLs).²

图表 3: Iran Exported 1.7mb/d of Crude and Condensates and 0.6mb/d of Refined Products in 2025



资料来源: Kpler, 高盛全球投资研究部

The Strait is crucial for nearly 20mb/d or 1/5 of global oil production (图表 1). Exports in 2025 through the Strait averaged 13.4mb/d for crude, 0.2mb/d for condensates, 3.5mb/d for refined products, and 1.4mb/d for NGLs and other liquids. Saudi Arabia, Iraq and the UAE together exported 13.1mb/d of oil via the Strait of Hormuz last year, with China as the main destination (图表 12).

While estimates vary across agencies, the IEA has estimated that 4.2mb/d of the oil flows through the Strait can be redirected using existing spare pipeline capacities, implying around 16mb/d of oil flows at risk from a hypothetical full closure of the

¹ That said, Iran's semi-official Mehr News Agency reported on Saturday an explosion at the Kharg Island oil export terminal, which handles over 90% of Iran crude exports. The Duqm port has also been targeted with no material damage reported.

² By refined product, Iran's 2025 exports stood at 0.3mb/d for fuel oil, 0.2mb/d for refined naphtha, and 0.1mb/d for other clean products.

Strait.³

Market Pricing

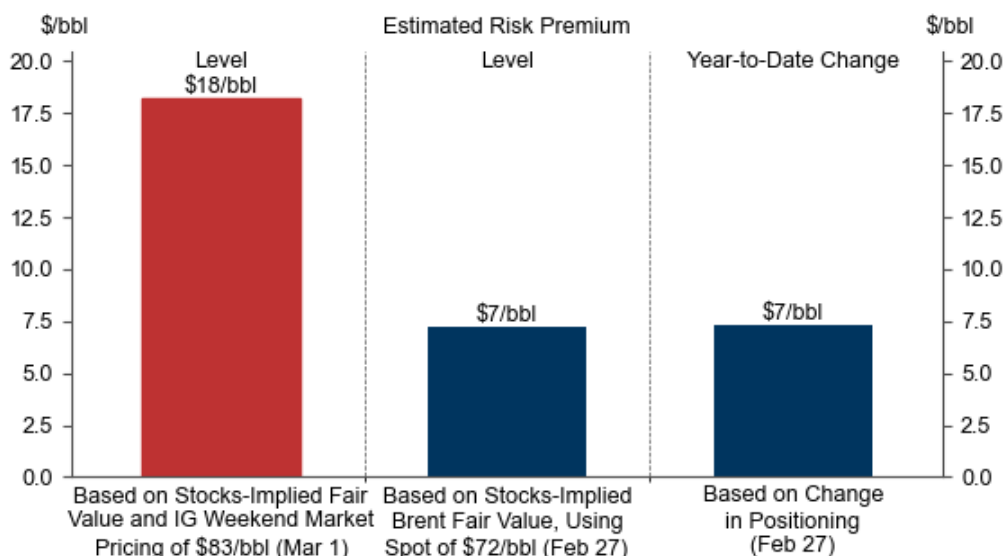
Q4. What risk premium is the oil market pricing?

We estimate the risk premium embedded in crude prices stood at \$7/bbl on Friday evening as the difference between the Brent Friday close of \$72 and our \$65 estimate of fair value based on fundamentals such as OECD stocks.

Based on the 15% gain in the WTI retail trading product, we estimate a real-time risk premium embedded in crude prices of \$18/bbl. This corresponds to the 98th percentile since 2005.

Call options offering upside protection embody a larger premium with 3 month 25-delta call skew at the 100th percentile of the distribution of the last 15 years already as of Friday.

图表 4: Taking Into Account Estimated Weekend Market Pricing, We Estimate a Risk Premium of Around \$18/bbl on the March 1st Oil Market Open



资料来源：IG, 高盛全球投资研究部

Q5. How does that real-time risk premium translate in oil supply disruptions?

The \$18/bbl real-time risk premium estimate corresponds to the market pricing a 1-year disruption of 2.3mb/d to global supply or approximately a 1-month full halt in Strait of Hormuz flows (though allowing for spare pipeline capacity use as a partial offset).⁴

³ The EIA estimates spare pipeline capacity that can bypass the Strait of Hormuz at 2.6mb/d while the IEA estimates are at 4.2mb/d. The main difference across estimates results from different assumptions on capacity at use and whether the Jask terminal and the Goreh-Jask pipeline are operating. Eurasia estimates pipeline bypass at 6.8mb/d (total) which is consistent with the IEA total pipeline capacity. CSIS estimates there is 3.4mb/d spare pipeline capacity. Finally, Kpler estimates that there is around 5mb/d of spare capacity in Saudi Arabia's East-West Pipeline alone, and 0.4-0.5mb/d of spare capacity in the UAE's ADCOP pipeline.

⁴ This uses our rule of thumb that a 1mb/d oil supply disruption for 1 year raises the fair value of oil

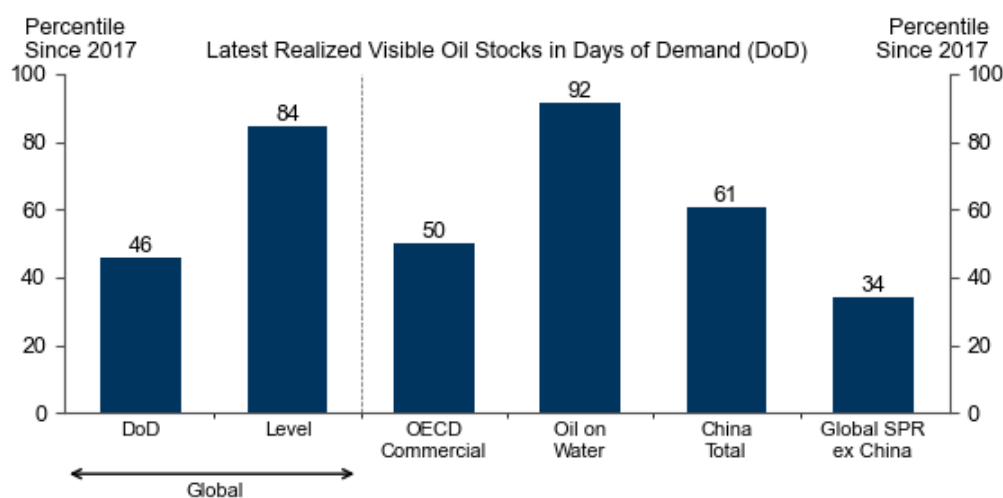
Scenarios

Q6. What buffers does the oil market have to absorb negative supply shocks? And what is OPEC+ signaling?

The oil market could draw inventories, deploy spare capacity once the Strait reopens, and potentially benefit from global SPR releases.

Global inventory levels rose last year from very low levels as global supply exceeded global demand by an estimated 1.7mb/d. Global total visible oil inventories stand at 7,827 million barrels now, near their historical median when expressed as covering 74 days of global demand (DoD). Relative to demand, inventory levels are also close to the median for OECD commercial stocks (the most predictive of prices), somewhat high in China, very high on water (with 373mb of sanctioned crude on water), but low in global Strategic Petroleum Reserves, excluding China⁵. The US SPR consists of 415mb (down around 180mb from end-2021). However, a US Department of Energy official is reported to have told the FT that there have been “no discussions at all about the SPR” , potentially signaling Washington believes any surge in prices will likely be limited or short-lived.

图表 5: Global Total Visible Oil Inventories as Days of Global Demand Are Now Close to the Historical Median



Stocks in DoD is calculated by dividing by 12MMA demand. All categories except for "Global SPR ex China" contain crude and products. Latest realized is Feb 26.

资料来源：IEA, Kpler, DOE, Euroilstocks, ARA PJK, PAJ, Haver, 高盛全球投资研究部

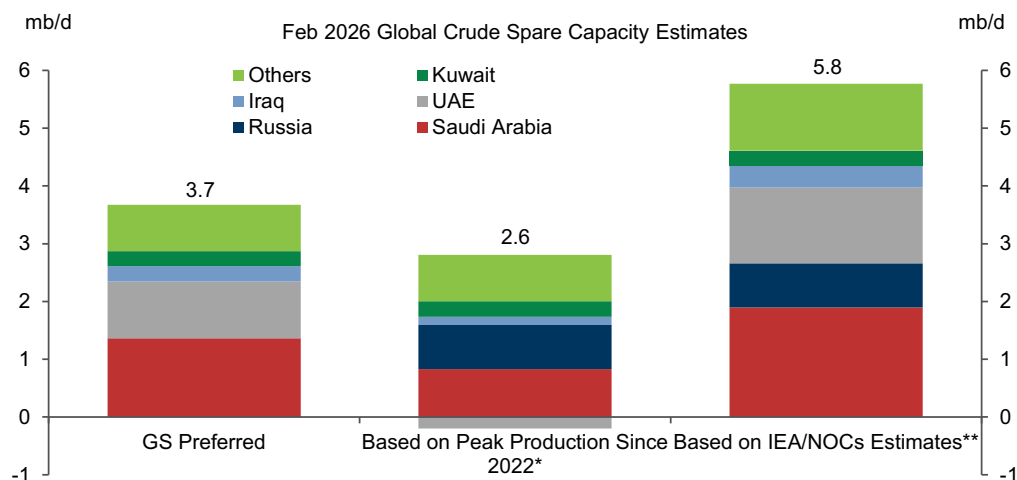
While highly uncertain, our preferred point estimate for global spare capacity is 3.7mb/d, largely concentrated in Saudi Arabia and the UAE (图表 6). A hypothetical sustained closure of the Strait of Hormuz would be a physical impediment to OPEC' s ability to deploy this spare capacity. Today, OPEC+ also announced it would raise required production by 0.21mb/d in April, slightly above our 0.14mb/d expectation.

prices by \$8/bbl.

⁵ We exclude China from our Global SPR estimates, where the distinction between government-owned and commercial inventories is nuanced.

While US shale remains the marginal short-term producer, US production has become less price-sensitive giving maturing shale, and raising US production significantly typically takes a few quarters.⁶

图表 6: While Highly Uncertain, Our Preferred Point Estimate of Global Spare Capacity Is 3.7mb/d



We define spare capacity as production that can be brought online within 30 days and sustained for at least 90 days. *Country estimate=Peak crude production since 2022 based on 3 month moving averages (3MMA) – Feb 2026 GS estimate. The Saudi Arabia and the UAE peak 3MMA production estimate are 10.9mb/d (Oct. 2022, OPEC secondary sources), and 3.3mb/d (Oct. 2022, GS Balance). **Country estimate= Full crude potential (based on IEA/NOCs) – Feb 2026 GS crude estimate. Saudi Arabia and UAE full crude potential production estimates from the NOCs are 12mb/d (Saudi Aramco) and 4.85mb/d (ADNOC), respectively.

资料来源：高盛全球投资研究部

Q7. How much could the fair value of oil prices rise in key disruption scenarios?

We think of Iran-related oil disruption scenarios as either reductions in Iran oil supply only, or disruption to broader regional oil shipping or production.

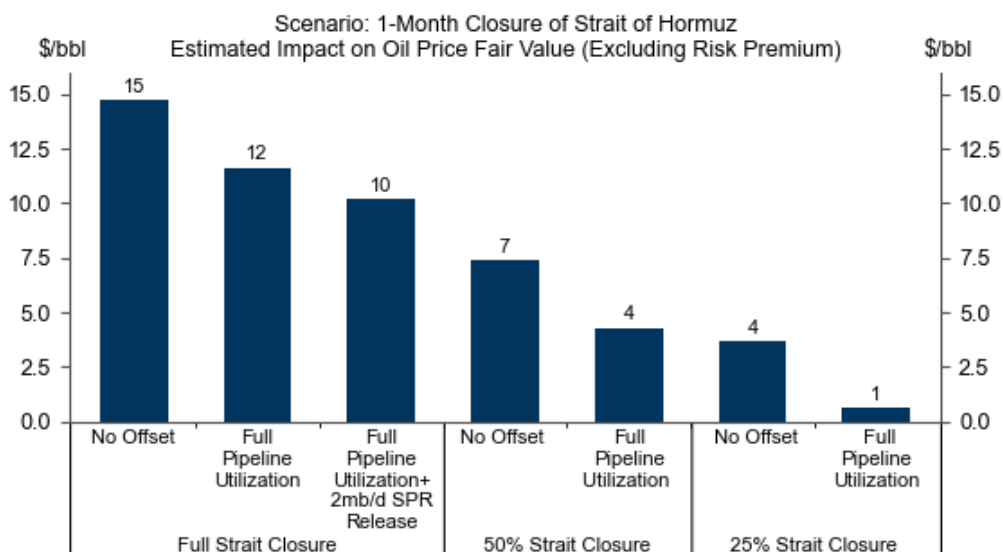
- Iran production disruptions: We estimate that a potential 1mb/d supply disruption—which corresponds to half of Iran’s crude exports—for 12 months would boost the fair value of oil by \$8 (although prices could rise more as escalation risks are reassessed).
- Broader production disruptions: Because Iran targeted oil producers across the Middle East, we can’t rule out future damage to oil production/export infrastructure in the region. The most recent comparable event is Iran’s attack in September 2019 on the Saudi Abqaiq oil production facility, which caused the biggest ever single day oil production disruption and a nearly 20% intraday price move (although prices moderated as production fully recovered just over 10 days after the initial strike (图表 9)).
- Hormuz shipping disruptions: While the actual price effects could become more extreme if risk premia rise, we next estimate the following effects on the fair

⁶ We estimate that US shale oil production growth accelerates (vs. the underlying trend) by 0.2mb/d over 12 months per \$10/bbl price increase with Brent >\$70/bbl, with this boost rising to 0.5mb/d per \$10 with Brent at \$50-70.

value of oil prices in scenarios for one-month disruptions to oil flows through the Strait:

- +\$15 for a full one-month closure if there are no offsets (e.g. utilization of spare pipeline capacity, SPR release)
- +\$12 for a full one-month closure if all estimated 4mb/d spare pipeline capacity is used
- +\$10 for a full one-month closure if all estimated spare pipeline capacity is used and global SPRs are released for one month at a 2mb/d pace
- +\$4 for a partial 50% one-month closure if all estimated spare pipeline capacity is used
- +\$1 for a partial 25% one-month closure if all estimated spare pipeline capacity is used

图表 7: We Estimate the Fair Value of Oil Prices Can Rise by 10-15 in a One Month Full Strait of Hormuz Closure Scenario The Estimated Impact on Oil Prices From Strait of Hormuz Disruptions Depends on the Disruption Length and Size and on the Use of Offsets



We do not take into account any potential boost to long-dated prices from a potential perceived reduction in medium-term spare capacity.

资料来源：高盛全球投资研究部

Can we simply scale the estimated fair-value effects for a month disruption from 图表 5 by 0.25 and by 4 if the disruption were to last $\frac{1}{4}$ of a month or 4 months?

No, while our simple static model directly relates net exports from the Middle East to global and OECD inventories, the actual fair-value effects likely rise disproportionately with the length of the disruption.

If the disruption lasted $\frac{1}{4}$ of a month, the hit to production would likely be very small as producers would likely continue to store oil on land, and delay rather than significantly cut cumulative exports. Moreover, producers such as Saudi Arabia have precautiously increased exports and production ahead of the escalation.

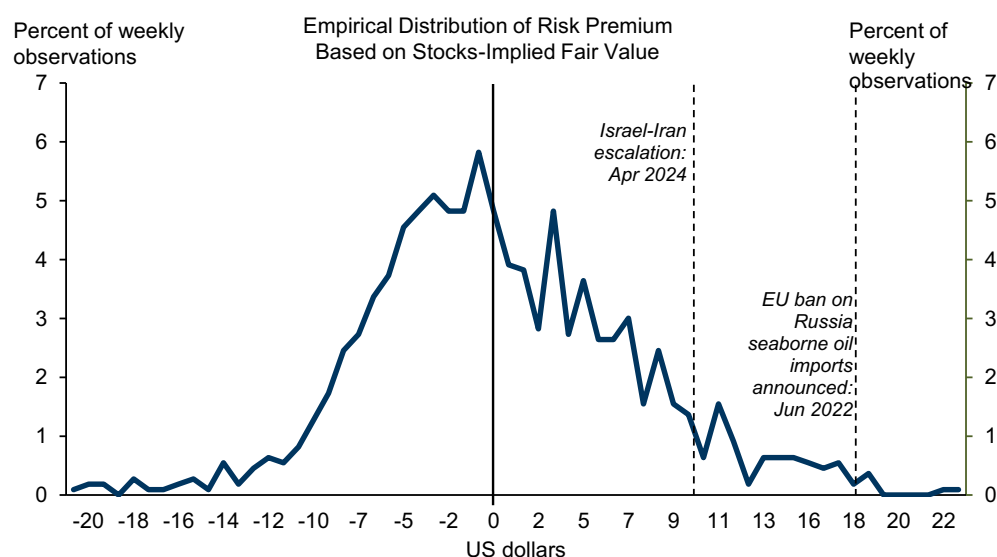
In contrast, if oil were to be trapped for four months in the region and because inventories can't draw persistently, oil prices would likely rise disproportionately to balance the market via price-driven demand destruction and refinery run cuts.

Q8. How much higher could risk premia lift oil prices?

While the scale and length of any potential disruption to oil flows from the Middle East and via the Strait of Hormuz remain uncertain, we have not changed our baseline oil price forecast for now, and still expect Brent/WTI prices to decrease to \$60/56 by 2026Q4 (assuming no sustained supply disruptions). While the risks to our forecast are skewed to the upside, history suggests that price spikes driven by geopolitical shocks or/and temporary supply disruptions can be short-lived (图表 9).

History shows that oil prices can rise significantly, and well above fair-value estimates when geopolitical uncertainty is high and when the market puts some weight on supply disruptions persisting. For instance, in mid-2022 oil prices exceeded our stocks-implied fair value estimate by nearly \$20 (图表 8).⁷ And Brent rose from the mid-60s in early June 2025 to the low \$80s when Israel and the US struck Iran nuclear facilities before retracing quickly when the market gained confidence that actual oil supply was unlikely to be disrupted (图表 9, right panel).

图表 8: History Shows That Oil Prices Can Rise Significantly Above Fair Value Estimates

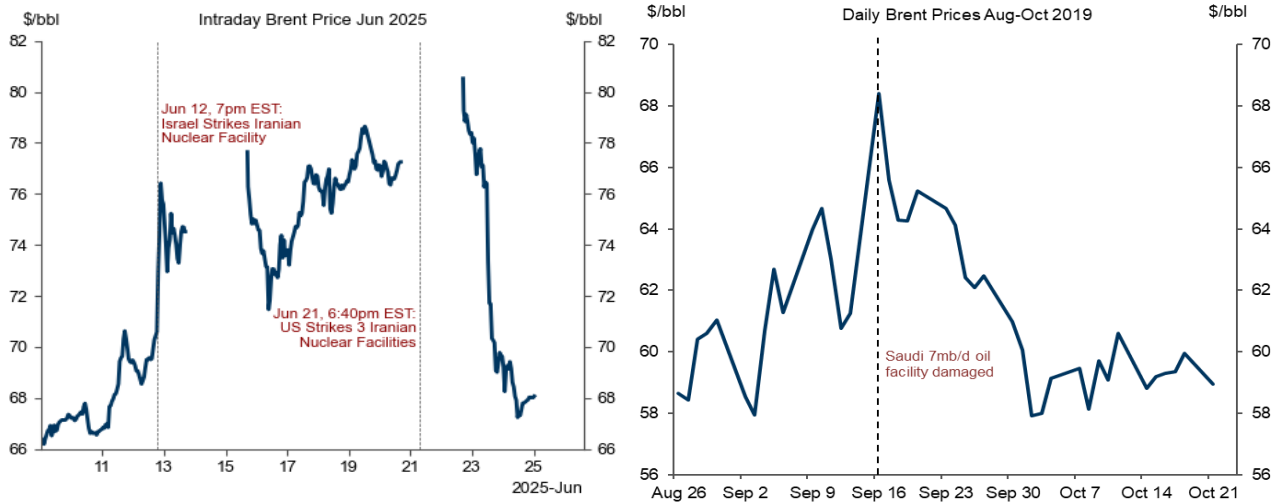


We estimate the empirical distribution of weekly risk premium as the difference between the Brent 1M/36M timespread and its inventory implied fair value, multiplied by the corresponding week's Brent 36M futures price. The sample includes observations from Jan 2005 till the latest.

资料来源：高盛全球投资研究部

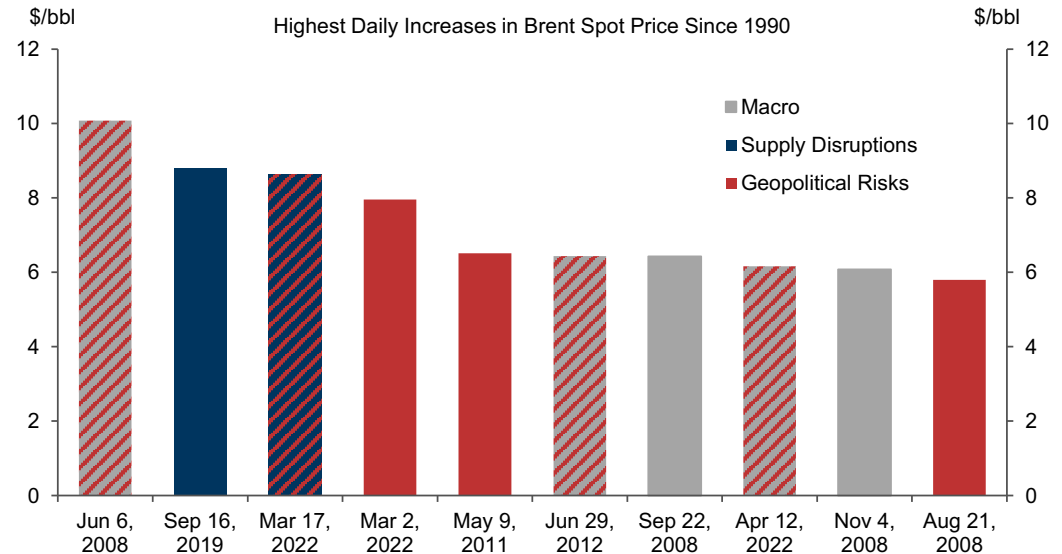
⁷ The percentiles 99 and 100 of our risk premium are \$21 and \$27/bbl.

图表 9: Oil Prices Can Normalize Quite Fast After Short-Lived Geopolitical Shocks and/or Temporary Supply Disruptions



资料来源：高盛全球投资研究部

图表 10: Largest Daily Brent Price Moves Are Typically Driven by Supply Disruptions and/or Rises in the Geopolitical Risk Premium



资料来源：高盛全球投资研究部

Q9. What is the potential impact on refined products and tanker freight markets?

The impact on refined products markets from any potential disruption in flows from Iran only would likely be limited given that Iran exports only 0.5mb/d of refined products.

But disruptions in the Strait of Hormuz flows could be very significant for middle distillates (diesel, gasoil), jet fuel, and naphtha markets. Last year, roughly 9% (or 0.8mb/d) of gasoil and diesel global exports went through the Strait of Hormuz and 18% of global jet fuel exports (0.4mb/d), which implies further upside risks to our already constructive refined products margins forecast, especially in Asia.

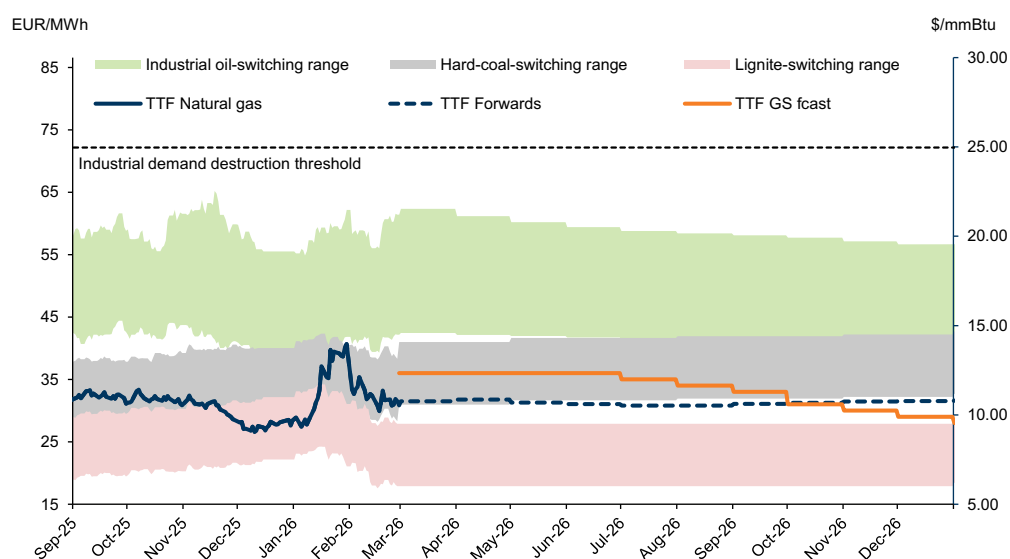
The escalation in the Middle East also poses substantial upside risks to tanker freight markets as insurance premia can rise very sharply. Our average global dirty (crude) freight rate has already increased by 50% (\$2.4/bbl) year-to-date as (mostly sanctioned) oil on water continues to build and Venezuela pivots from the dark fleet to non-sanctioned vessels. The very Large Crude Carrier (VLCC) freight rate from the Arab Gulf to China had already tripled over the last month on Friday's close. And even without significant further disruptions in the Strait, precautionary restocking and redirection can raise already elevated freight rates further.

Natural Gas

Q10. How much risk premium has been embedded in European gas prices?

Differently from oil, we believe that until this past Friday, European natural gas prices had embedded little-to-no risk premium associated with Iran-related geopolitical risks. Specifically, TTF has been pricing in the bottom half of our estimated hard-coal-to-gas (C2G) switching range for the past month, modestly below our 36 EUR/MWh March 2026 TTF forecast (图表 11). Once accounting for the recent sell off in carbon emission prices to below the 80 EUR/t embedded in our TTF price forecast, worth 2.0-2.5 EUR/MWh in gas-equivalent terms, prompt gas prices are still largely in line with our view that TTF needs to be in this C2G switching range to help manage NW European gas storage to above 80% full by end-Oct26, given that current inventory levels remain well below average. That remains our view, and we maintain our 34 EUR/MWh balance-of-the-year TTF price forecast.

图表 11: TTF prices have been in the lower end of the hard coal switching range, in line with our 34 EUR/MWh average 2026 forecast
Realized and forward TTF prices and fuel switching ranges, and TTF GS fcast



Prices shown in gas-equivalent terms, adjusted for carbon costs and burning efficiency; oil products adjusted for carbon costs only.

资料来源: ICE, S&P Global Commodity Insights, 高盛全球投资研究部

Q11. What are the risks to global gas prices from this weekend's developments in the Middle East?

We see significant upside risk to European gas and global LNG prices. The most significant impact to global gas markets would come from a potential disruption of the approximately 80 mtpa (302 mcm/d or 11 Bcf/d, 19% of global LNG supply) of LNG that typically flow through the Strait of Hormuz (图表 2), which could potentially arise from an escalation of the ongoing conflict.

Specifically, in a scenario where LNG flows through the Strait are fully halted for one month, we estimate a resulting tightening of NW European gas storage equivalent to

8% of capacity⁸. Our fuel switching models suggest that European gas prices would need to maximize both switching into hard coal and into oil products by pricing at or above distillate fuel price levels for over three and a half months to offset it⁹. At current oil prices this would imply TTF essentially doubling to 62 EUR/MWh¹⁰ (\$21/mmBtu). Given that oil prices would also likely rally in this scenario, it is likely that TTF could approach the 74 EUR/MWh (\$25/mmBtu) threshold that triggered large natural gas demand responses during the 2022 European energy crisis.

A hypothetical longer disruption of natural gas supply transit through the Strait of Hormuz lasting more than two months would likely lift European natural gas prices above 100 EUR/MWh (\$35/mmBtu) to trigger more significant global gas demand destruction given the increased difficulty for the market to fully offset such a tightening shock ahead of the next winter.

Given that spot LNG prices (JKM) are closely tied to TTF to help Asia manage Atlantic-basin-produced LNG flows to the Pacific basin to satisfy its demand, the upside risks to TTF discussed above also apply to JKM.

Q12. Are there risks to global gas markets from Iranian pipeline gas exports?

We think these risks are very limited, as the global gas market exposure to Iran's pipeline gas exports is modest, with Iran's flows having decreased in recent years owing primarily to a growing domestic deficit. The most relevant remaining indirect link to global gas markets are its exports to Türkiye, which averaged just over 8 Bcm in 2025. These flows are under a contract that expires this summer and has yet to be renewed.

A scenario in which these flows are halted might trigger an increase in Türkiye LNG imports of similar size and, with NW Europe typically taking about half of total European LNG imports, might tighten NW European gas balances by the equivalent of half of the net 8 Bcm/y potential increase in Türkiye LNG imports, or 4 Bcm/y (11 mcm/d or 0.4 Bcf/d). To remind, we focus on NW Europe specifically (UK, Belgium, France, Netherlands and Germany) because we see it as the most relevant region to set TTF prices given its pipeline connectivity. We estimate that TTF would need to rally 4 EUR/MWh relative to hard coal prices to offset such tightening. This would justify TTF moving from the low end of the hard-coal switching range, where it has been for the past four weeks, to the high end, near 36 EUR/MWh.

Q13. What are the risks to US Henry Hub gas prices?

⁸ A potential Strait of Hormuz LNG flow disruption worth 80 mtpa or 302 mcm/d could tighten NW European balances by about 50% of the total LNG disruption (given NW Europe typically takes 50% of total European LNG imports), or about 151 mcm/d. Over 31 days, that's 4.68 Bcm, equivalent to 8.1% of NW European gas storage capacity.

⁹ We estimate that NW European gas-to-coal switching can increase another 19 mcm/d to a max of 27 mcm/d by TTF pricing at approximately 39 EUR/MWh. Further, we estimate that gas-to-oil switching can reach 24 mcm/d based on what we observed during the 2022 European energy crisis, for a total of 43 mcm/d of potential gas demand destruction if TTF prices can be sustained at or above distillate fuel prices.

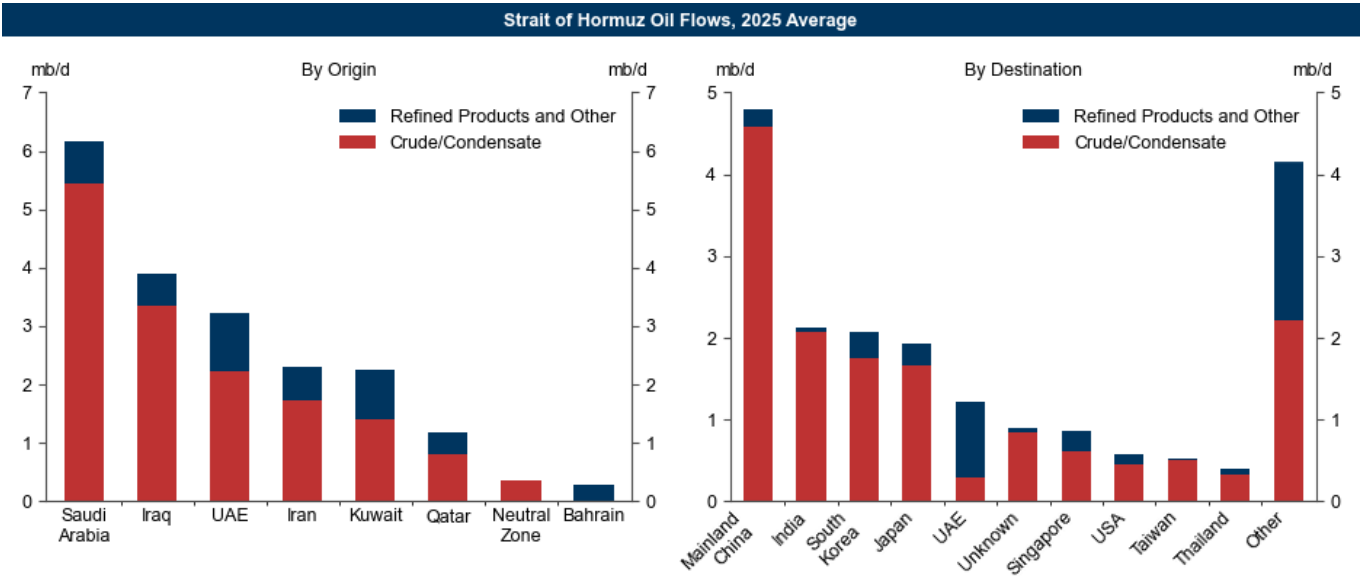
¹⁰ This cost is adjusted for heating content and carbon cost differentials.

Under a scenario of a large global LNG supply disruption, we believe the impact to US natural gas balances would be limited for three main reasons. First, the US is a large net exporter of LNG, with importing needs limited to very few cargoes in the US Northeast coast during cold spikes in winter. Second, Mexico and Canada, connected to US gas markets via pipelines, also have very limited LNG importing needs. Third, owing to low variable costs, US LNG export facilities typically operate at capacity, only reducing utilization rates during maintenance periods. As a result, there is little to no room for US LNG exports to rise even in the event of a global LNG price spike.

While US gas balances might tighten at the margin if oil product shortages regionally in the US or in Mexico lead to an increase in oil-to-gas switching for power generation or industrial applications, we think that upside risks to Henry Hub are limited. To be clear, differently from the tight 1H2022 US gas balance that caused Henry Hub to rally with Appalachia coal prices when international coal prices spiked with natural gas at the time, current US gas balances are more comfortable, as reflected in prompt US gas prices below \$3/mmBtu and a very negative Oct26-Jan27 spread, signaling a market perception of elevated storage congestion risks this year.

Appendix

图表 12: Saudi Arabia, Iraq and the UAE Together Exported 13.1mb/d of Oil via the Strait of Hormuz Last Year, With China as the Main Destination



资料来源: Kpler, 高盛全球投资研究部

信息披露附录

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